Assessment of Landslide Impact on Vegetation

Using NDVI from Landsat Satellite Images

A Case Study of Mount Rinjani, Indonesia

Study Period: April 2018 – October 2025
Study Sites: Carbon Valley Gully & Paradise Valley Gully
Data Source: Landsat 8/9 OLI via Google Earth Engine

Abstract

This study quantifies the impact of the 2018 Lombok earthquake-triggered landslides on vegetation cover at Mount Rinjani, Indonesia, using Normalised Difference Vegetation Index (NDVI) derived from Landsat 8/9 OLI satellite imagery across the period April 2018 to October 2025. Two gully sites — Carbon Valley Gully (26 sample points) and Paradise Valley Gully (24 sample points) were analysed across growing (wet, November–April) and non-growing (dry, May–October) seasons. Results show that the August 2018 earthquake caused severe vegetation loss of 80.97% at Carbon Valley Gully and 93.81% at Paradise Valley Gully relative to the April–May 2018 pre-earthquake baseline. Post-disturbance recovery was nonlinear, with secondary slumps in 2020 and 2022 followed by a marked recovery in 2023 (peak mean NDVI: Carbon 0.867, Paradise 0.880 in July 2023). Inter-site Pearson correlation ($r = 0.899$, $p < 0.001$) confirms that both valleys respond synchronously to the same macro-scale disturbance and recovery drivers. Mann-Whitney U tests demonstrate significantly higher NDVI during the dry season at both sites ($p < 0.001$), attributable to cloud cover reducing optical satellite signal quality during the monsoon. These findings highlight the severity and spatial variability of earthquake-induced landslide disturbance and the long, nonlinear trajectory of tropical montane forest recovery.

Keywords: NDVI, landslide, vegetation recovery, Mount Rinjani, Landsat, earthquake, remote sensing, Indonesia

1. Introduction

Natural disturbances such as earthquake-triggered landslides can cause catastrophic vegetation loss across tropical mountain landscapes. Mount Rinjani, the second highest volcano in Indonesia (3,726 m a.s.l.), located on the island of Lombok, West Nusa Tenggara, experienced a sequence of powerful earthquakes in July and August 2018. The main shock of moment magnitude (Mw) 6.9 on 5 August 2018 triggered widespread slope failures on the mountain's flanks, stripping vegetation, exposing bare soil, and dramatically altering the local ecological landscape.

Satellite remote sensing provides an efficient and reproducible means to monitor vegetation change over large, difficult-to-access areas. The Normalised Difference Vegetation Index (NDVI), derived from the near-infrared (NIR) and red spectral bands of Landsat imagery, is widely used as a proxy for above-ground vegetation density and health. NDVI ranges from −1 to +1, with values above 0.5 generally indicative of dense, healthy vegetation canopy.

This study addresses three principal objectives:

1. To quantify the extent of vegetation loss caused by the 2018 earthquake-induced landslides at two gully sites on Mt. Rinjani.
2. To assess post-disturbance vegetation recovery trajectories across growing and non-growing seasons from 2018 to 2025.
3. To evaluate inter-annual NDVI relationships to identify spatial and temporal patterns of vegetation change.

2. Study Area

The study focuses on two gully systems on the southern and south-eastern flanks of Mt. Rinjani, identified through spatial clustering of 50 satellite-derived sample points:

Site	Coordinates (centre)	Sample Points
Carbon Valley Gully	8.477°S, 116.417°E	n = 26
Paradise Valley Gully	8.477°S, 116.439°E	n = 24

Both sites are situated within the Mt. Rinjani National Park, characterised by tropical montane forest at elevations between approximately 800 and 1,500 m a.s.l. The region experiences the Indonesian monsoon, with a pronounced wet season from November to April and a dry season from May to October. Annual rainfall typically exceeds 2,000 mm, supporting dense evergreen forest cover under undisturbed conditions.

3. Data and Methods

3.1 Satellite Data

NDVI was extracted from Landsat 8 and Landsat 9 Operational Land Imager (OLI) surface reflectance imagery processed through Google Earth Engine. A total of 42 observation dates were obtained between April 2018 and October 2025 (Table 3). Only cloud-free or near-cloud-free scenes were retained. NDVI values falling outside the physically valid range of −1 to +1, attributable to residual cloud contamination or sensor artefacts, were excluded from analysis (n = 46 values set to missing; no interpolation was applied).

3.2 Spatial Site Assignment

The 50 sample points were assigned to two valley units using k-means clustering (k = 2) applied to the geographic coordinates (longitude and latitude). Cluster assignment was made robust by anchoring labels to the cluster centroid with lower longitude, corresponding to the westward Carbon Valley Gully. This approach was used in preference to a simple threshold split to account for the irregular spatial distribution of sampling points.

3.3 Seasonal Definition

Seasons were defined in accordance with the Indonesian monsoon calendar. The growing (wet) season spans November to April, corresponding to periods of active vegetation growth facilitated by monsoon rainfall. The non-growing (dry) season spans May to October. It is important to note that dry-season observations tend to yield higher apparent NDVI values at both sites, attributable primarily to reduced cloud cover and associated lower optical noise in Landsat imagery during the dry monsoon period rather than to genuinely higher vegetation density.

3.4 Vegetation Loss Calculation

Pre-earthquake baseline NDVI was established as the mean of all April and May 2018 observations — the last available cloud-free observations prior to the onset of the earthquake sequence. The July 2018 acquisition was excluded from the baseline because early foreshocks beginning on 29 July 2018 had already initiated slope instability. Post-disturbance NDVI was taken as the mean of August and September 2018 observations. Vegetation loss was computed as:

$$\text{Vegetation Loss (\%)} = [(\text{NDVI_pre} - \text{NDVI_post}) / \text{NDVI_pre}] \times 100$$

3.5 Statistical Analysis

Inter-site Pearson correlation was computed between the monthly mean NDVI time series of Carbon Valley Gully and Paradise Valley Gully to assess co-movement. Mann-Whitney U tests were applied independently at each site to test whether mean NDVI differed significantly between growing and non-growing seasons. All analyses were implemented in Python 3 using pandas, NumPy, scikit-learn, and SciPy.

4. Results

4.1 Objective 1: Vegetation Loss from Earthquake-Induced Landslides

Pre-earthquake NDVI (April–May 2018) indicated dense, healthy vegetation at both sites: Carbon Valley Gully (mean = 0.709 ± 0.18) and Paradise Valley Gully (mean = 0.811 ± 0.09). Following the August 2018 earthquake and associated landslides, NDVI collapsed to near-zero at both sites by August–September 2018 (Carbon: 0.135; Paradise: 0.050), representing vegetation losses of 80.97% and 93.81% respectively (Table 1 context; Figure 1).

Paradise Valley Gully experienced substantially greater relative loss, suggesting a more severe or more extensive landslide footprint at that site. The higher pre-earthquake baseline NDVI at Paradise Valley (0.811 vs 0.709) indicates denser pre-disturbance canopy, which may have contributed to the proportionally greater detectable loss.

Table 1. Seasonal NDVI Summary Statistics — Both Study Sites (2018–2025)

Metric	Carbon Valley Gully	Paradise Valley Gully
Growing season mean NDVI	0.270	0.356
Growing season SD	0.329	0.351
Growing season minimum	−0.197	−0.015
Growing season maximum	0.999	0.927
Non-growing season mean NDVI	0.487	0.511
Non-growing season SD	0.348	0.345
Non-growing season minimum	−0.009	−0.003
Non-growing season maximum	0.952	0.964

Values represent means and standard deviations across all observations within each season category. Both sites show significantly higher NDVI during the non-growing (dry) season.

4.2 Objective 2: Post-Disturbance Recovery Trajectories

Post-earthquake recovery was nonlinear and characterised by two distinct phases (Figure 1; Table 2). An initial partial recovery was observed by July 2019, with both sites returning to NDVI values approximately 0.70, suggesting rapid pioneer vegetation colonisation of disturbed slopes within approximately 11 months of the earthquake.

However, a secondary decline occurred in 2020 (annual mean NDVI: Carbon 0.013, Paradise 0.033 — the lowest post-disturbance values recorded), followed by further suppression in 2022 (Carbon 0.160, Paradise 0.068). The 2020 decline coincides with the global disruption of cloud-free satellite acquisition opportunities during the COVID-19 period and may also reflect secondary slope instability or volcanic aerosol effects associated with Rinjani's ongoing fumarolic activity.

A sustained recovery phase commenced in 2023, when both sites recorded their highest post-earthquake NDVI values (July 2023: Carbon 0.867, Paradise 0.880). By 2025, NDVI at both sites (Carbon 0.488, Paradise 0.566 annual mean) had broadly recovered toward pre-earthquake levels, though full canopy recovery remains incomplete at some pixel locations, particularly those experiencing persistent bare-soil or slow-colonising coverage.

Table 2. Annual Mean NDVI — Carbon Valley Gully and Paradise Valley Gully (2018–2025)

Year	Carbon Valley Gully	Paradise Valley Gully
2018	0.468	0.454
2019	0.399	0.459
2020	**0.013**	**0.033**
2021	0.474	0.525
2022	0.160	**0.068**
2023	**0.633**	**0.700**
2024	0.373	0.446
2025	0.488	**0.566**

Green = strong recovery (mean > 0.55); Red = severe degradation (mean < 0.10). Values are means across all observations within each calendar year.

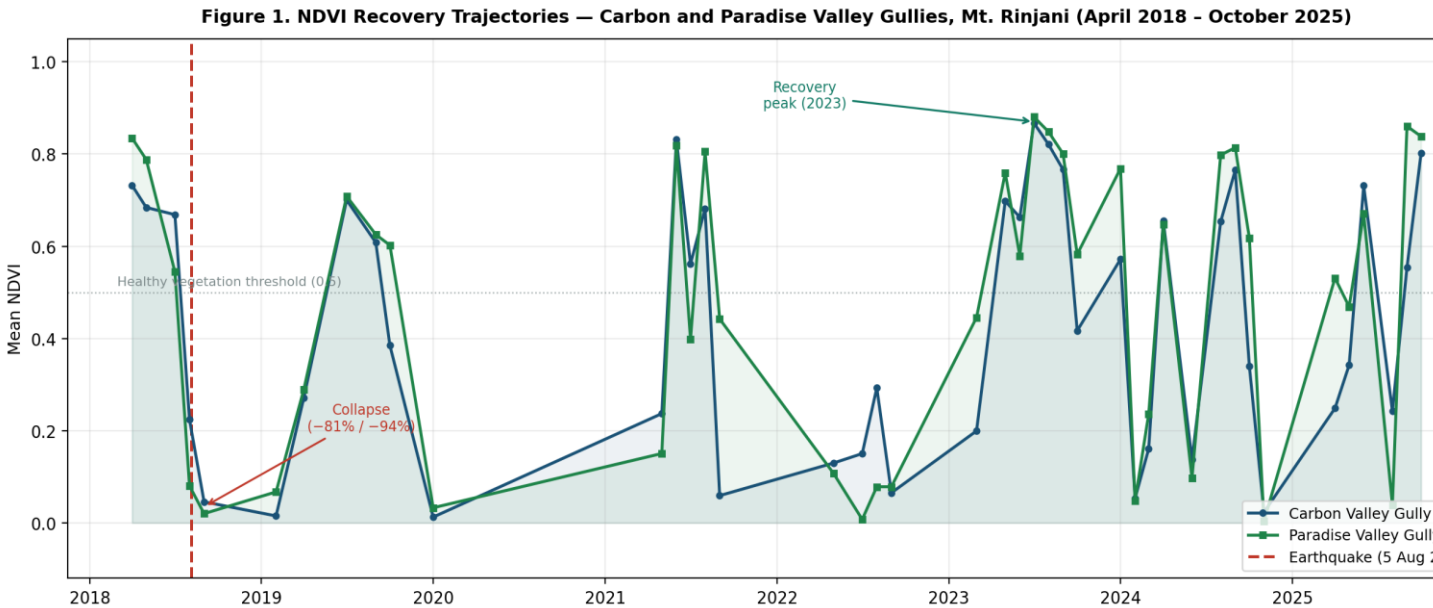

Figure 1. Monthly mean NDVI time series for Carbon Valley Gully and Paradise Valley Gully, April 2018 – October 2025. The dashed red vertical line marks the 5 August 2018 earthquake (Mw 6.9). The horizontal dotted line at NDVI = 0.5 represents the general threshold for healthy dense vegetation. Shaded areas show ±1 standard deviation across individual sample points.

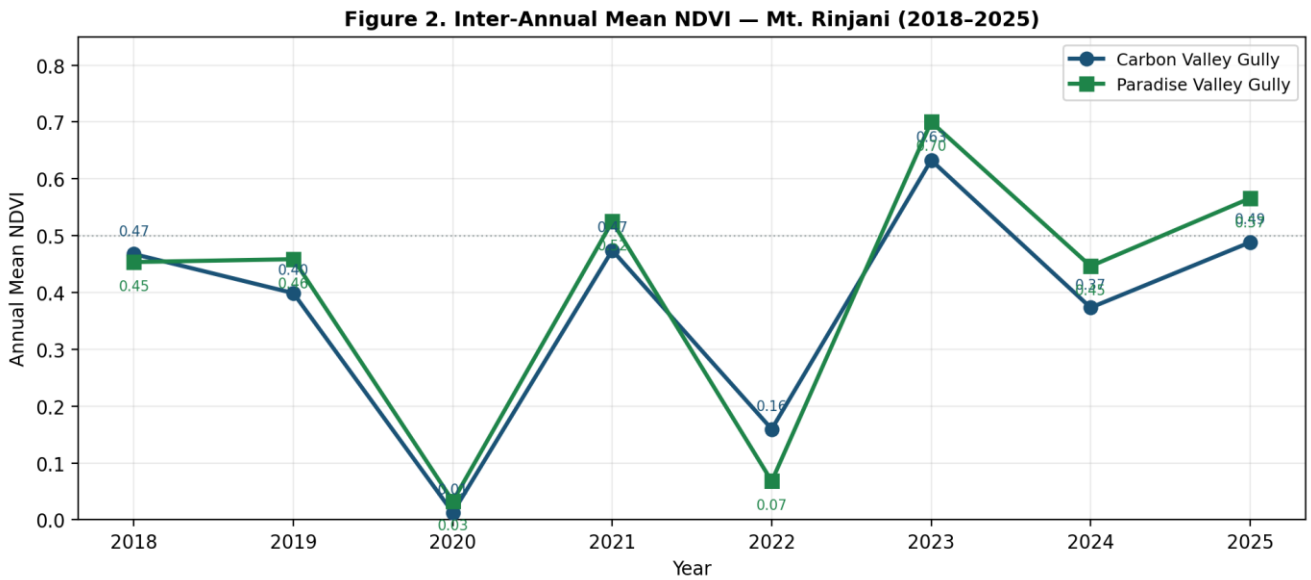

Figure 2. Annual mean NDVI for Carbon Valley Gully (blue) and Paradise Valley Gully (orange), 2018–2025. Values annotated above (Carbon) and below (Paradise) each data point. The 2020 and 2022 secondary slumps and the 2023 recovery breakthrough are clearly visible in both series.

4.3 Objective 3: Inter-Annual Relationships and Statistical Patterns

The Pearson correlation between the monthly mean NDVI series of Carbon Valley Gully and Paradise Valley Gully was $r = 0.899$ ($p < 0.001$, $n = 40$ paired observations). This strong positive correlation confirms that both sites respond synchronously to the same macro-scale drivers: the 2018 earthquake disturbance, regional climate variability, and the broader recovery trajectory shared across the Mt. Rinjani landscape. The positive relationship holds across all phases — disturbance, initial recovery, secondary decline, and sustained recovery — indicating that site-level differences in recovery rate are modulated by local factors (slope aspect, landslide scar extent, soil depth) rather than fundamentally different ecological trajectories.

Mann-Whitney U tests demonstrated significantly higher NDVI during the non-growing (dry) season at both sites (Carbon Valley Gully: $U = 70{,}079$, $p < 0.001$; Paradise Valley Gully: $U = 71{,}931$, $p < 0.001$). This seasonal asymmetry is primarily a measurement artefact: cloud cover during the wet monsoon (November–April) reduces the quality and availability of optical satellite imagery, resulting in apparently lower NDVI values during the growing season. The pattern does not imply that vegetation is genuinely less healthy during the wet season; rather, it underscores the importance of prioritising dry-season observations for vegetation monitoring in this climatic setting.

Table 3. Statistical Test Results — Mt. Rinjani NDVI Analysis

Test	Site	Statistic	Interpretation
Pearson Correlation	Both sites	**$r = 0.899$, $p < 0.001$**	Strong positive — both valleys respond to the same disturbance and recovery events
Mann-Whitney U	Carbon Valley Gully	**$U = 70{,}079$, $p < 0.001$**	Significantly higher NDVI in the dry (non-growing) season

Test	Site	Statistic	Interpretation
Mann-Whitney U	Paradise Valley Gully	**U = 71,931, p < 0.001**	Significantly higher NDVI in the dry (non-growing) season

5. Discussion

The vegetation loss estimates of 80.97% (Carbon Valley Gully) and 93.81% (Paradise Valley Gully) are consistent with the catastrophic nature of earthquake-triggered landslides documented in similar tropical montane settings. The greater relative loss at Paradise Valley Gully aligns with the higher pre-earthquake NDVI at that site, which may indicate a more mature or denser forest canopy that was therefore more detectable before disturbance.

The nonlinear recovery trajectory — initial partial recovery (2019), secondary decline (2020, 2022), breakthrough recovery (2023) — is characteristic of tropical forest post-disturbance dynamics. Secondary declines may reflect several interacting processes: continued slope instability following the primary earthquake, volcanic activity associated with Rinjani, inter-annual drought stress, and the heterogeneous nature of landslide scars which include areas of rapid pioneer colonisation alongside areas of persistent bare soil or debris.

The consistently higher dry-season NDVI observed at both sites should be interpreted cautiously. While it might initially suggest seasonally constrained vegetation productivity, the most parsimonious explanation is reduced cloud contamination in dry-season imagery, which allows more satellite observations to achieve the signal quality required for reliable NDVI extraction. Future analyses would benefit from cloud probability masking at the individual pixel level to separate genuine phenological cycles from measurement bias.

The strong inter-site correlation ($r = 0.899$) has practical implications for monitoring design: it suggests that the two gully systems can serve as mutual validation sites, and that conclusions drawn from one site are likely transferable to the other at the landscape scale. Local divergences — particularly the deeper 2022 slump at Paradise Valley — warrant targeted field investigation to identify site-specific drivers.

6. Conclusion

This study demonstrates that the 2018 Lombok earthquake caused catastrophic vegetation loss of 81–94% at two gully sites on Mt. Rinjani. Recovery over the subsequent seven years was nonlinear, with a breakthrough recovery phase beginning in 2023 bringing NDVI values close to pre-earthquake levels. The dry season consistently provides the most reliable satellite-derived vegetation signal in this monsoon environment, and both sites exhibit strongly correlated temporal dynamics driven by the same landscape-scale disturbance and recovery processes.

These findings contribute to the growing body of evidence on the long-term ecological consequences of earthquake-triggered landslides in tropical montane forests and have direct relevance for post-disaster ecological monitoring and restoration planning in the Mt. Rinjani National Park.

Continued annual satellite monitoring through 2026 and beyond is recommended to confirm whether recovery is sustained and to identify any persistent landslide scar areas requiring active ecological restoration intervention.

Appendix: Table 4 — Full NDVI Observation Record (April 2018 – October 2025)

Table 4. Mean NDVI per observation date for Carbon Valley Gully and Paradise Valley Gully. C. = Carbon Valley Gully; P. = Paradise Valley Gully. Green shading = pre-earthquake baseline (April–May 2018); Red shading = post-earthquake disturbance (August–September 2018).

Date	Season	C. Mean	C. SD	P. Mean	P. SD
2018-04	Growing (Wet)	0.732	0.174	0.835	0.033
2018-05	Non-Growing (Dry)	0.684	0.255	0.788	0.158
2018-07	Non-Growing (Dry)	0.668	0.287	0.545	0.230
2018-08	Non-Growing (Dry)	0.224	0.253	0.080	0.056
2018-09	Non-Growing (Dry)	0.046	0.155	0.020	0.031
2019-02	Growing (Wet)	0.015	0.023	0.067	0.076
2019-04	Growing (Wet)	0.271	0.275	0.289	0.271
2019-07	Non-Growing (Dry)	0.700	0.166	0.709	0.158
2019-09	Non-Growing (Dry)	0.608	0.143	0.625	0.132
2019-10	Non-Growing (Dry)	0.385	0.273	0.602	0.133
2020-01	Growing (Wet)	0.013	0.047	0.033	0.071
2021-05	Non-Growing (Dry)	0.237	0.256	0.151	0.173
2021-06	Non-Growing (Dry)	0.831	0.084	0.818	0.076
2021-07	Non-Growing (Dry)	0.562	0.185	0.398	0.201
2021-08	Non-Growing (Dry)	0.681	0.252	0.805	0.062
2021-09	Non-Growing (Dry)	0.059	0.127	0.442	0.308
2022-05	Non-Growing (Dry)	0.130	0.262	0.107	0.121
2022-07	Non-Growing (Dry)	0.150	0.249	0.008	0.004
2022-08	Non-Growing (Dry)	0.293	0.284	0.078	0.042
2022-09	Non-Growing (Dry)	0.065	0.104	0.078	0.093
2023-03	Growing (Wet)	0.200	0.288	0.445	0.368
2023-05	Non-Growing (Dry)	0.698	0.220	0.759	0.158
2023-06	Non-Growing (Dry)	0.663	0.246	0.579	0.243
2023-07	Non-Growing (Dry)	0.867	0.067	0.880	0.055
2023-08	Non-Growing (Dry)	0.821	0.088	0.848	0.063

Date	Season	C. Mean	C. SD	P. Mean	P. SD
2023-09	Non-Growing (Dry)	0.766	0.110	0.800	0.062
2023-10	Non-Growing (Dry)	0.417	0.372	0.582	0.249
2024-01	Growing (Wet)	0.572	0.418	0.769	0.233
2024-02	Growing (Wet)	0.053	0.066	0.048	0.049
2024-03	Growing (Wet)	0.161	0.216	0.237	0.239
2024-04	Growing (Wet)	0.656	0.073	0.648	0.077
2024-06	Non-Growing (Dry)	0.137	0.254	0.097	0.130
2024-08	Non-Growing (Dry)	0.653	0.289	0.798	0.111
2024-09	Non-Growing (Dry)	0.765	0.098	0.814	0.045
2024-10	Non-Growing (Dry)	0.339	0.315	0.618	0.242
2024-11	Growing (Wet)	0.023	0.035	0.004	0.017
2025-04	Growing (Wet)	0.249	0.262	0.531	0.281
2025-05	Non-Growing (Dry)	0.342	0.350	0.469	0.327
2025-06	Non-Growing (Dry)	0.732	0.177	0.671	0.277
2025-08	Non-Growing (Dry)	0.242	0.300	0.039	0.075
2025-09	Non-Growing (Dry)	0.554	0.302	0.860	0.033
2025-10	Non-Growing (Dry)	0.802	0.075	0.838	0.034

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